reciprocal exchange of advice. In the latter situation no differentiation of status occurs between participants, since advice received at one time is repaid not by deference but by giving advice at a later time. We may further explore these patterns by attempting to determine what the conditions are that lead to more or less reciprocity in a group's consultation network and by examining some implications of such reciprocity.

A measure of perceived reciprocity in consultation was obtained in City Agency by asking caseworkers which colleagues they consulted on work problems and which ones regularly consulted them. If several members of a work group named the same person on both questions, the group was assigned a high score on the reciprocity index; if not, it received a low score.[34] Our data suggest that work pressure promoted reciprocity in consultations. The indicator of work pressure used was the number of new cases assigned to a worker in a month, because new cases made urgent demands on a worker's time.[35] In work groups marked by high reciprocity, 13 out of 22 workers were working under high pressure (more than 15 new cases); in low-reciprocity groups only 4 out of 23 workers were under high pressure. This association suggests that workers under high pressure were more likely to develop reciprocal partnerships to meet their need for consultation, since such networks make it easier to obtain advice.

Once a reciprocal-consultation structure has developed in a group, it influences other aspects of the communication patterns. In high-reciprocity groups the amount of work pressure on individuals was directly associated with their tendency to consult colleagues,[36] but in low-reciprocity groups this association was inverse (see Table 11). Thus, in groups with reciprocal-consultation networks hardly any of the workers under little pressure (fewer than 11 new cases) regularly consulted colleagues, but the majority of those under much pressure did so. In contrast, in groups with little reciprocal consultation the

[34] The index of perceived reciprocity was computed by dividing the number of identical choices on both questions in a unit by the total number of choices made and received by workers in the unit. Units above the median were considered high in reciprocity. A measure of *actual* reciprocity in sociometric choices of consultants based on the choices of both partners was also computed. However, this index showed little relation to perceived reciprocity, and it was not associated with other factors in the same way as perceived reciprocity. Perhaps reciprocity is effective only if it is so perceived by the subjects; but it is also possible that the actual reciprocity index was unreliable, since there were very few actual reciprocated choices among workers.

[35] These data taken from the official records were available for only 45 of the 60 workers.

[36] Frequency of consulting others was determined by sociometric methods. Workers were asked which colleagues frequently came to them for advice. If a worker was named on this question by one or more respondents he was considered to consult others regularly.

TABLE 11. RECIPROCITY IN CONSULTATIONS, WORK PRESSURE, AND PER CENT REGULARLY CONSULTING COLLEAGUES

Number of New Cases (Work Pressure)	*Reciprocity in Group*	
	High	*Low*
10 or fewer (low)	17% (6)*	60% (5)
11 to 15 (medium)	100% (3)	57% (14)
16 and more (high)	54% (13)	0% (4)

* Numbers in parentheses are those on which corresponding percentages are based.

majority of the workers with few new cases asked for advice frequently, but none of those with many (more than 15) did so. This association suggests that the presence of reciprocal-consultation channels reduced the effort and social cost required for obtaining advice. In the absence of reciprocal partnerships, the greater the pressure of his work the less inclined a worker was to expend the extra effort necessary to get advice. In work groups where reciprocal networks had been established, on the other hand, consultants were readily available and the tendency to ask for advice was primarily governed by the need for it, which was greater if a worker had many new cases with new problems than if he had few. In short, reciprocal-consultation networks made advice readily available to workers when they needed it most and not merely when relatively low work pressure permitted them to devote special energy to obtaining it.

The extent of reciprocal consultation in a group not only influenced the conditions under which a worker asked for advice but also influenced whose advice was most often sought. In low-reciprocity groups, highly competent workers were more often consulted than others and were less likely to ask colleagues for advice.[37] But in high-reciprocity groups, there was no relationship between competence and either being consulted or consulting. This pattern is exactly what one would predict: workers in low-reciprocity groups typically go to experts for advice, since they have not established consultation partnerships, while workers in high-reciprocity groups do not concentrate their requests for advice on experts but usually consult in partnerships.

Reciprocity was also found to reverse the relationship between self-confidence and being consulted, as Table 12 shows.[38] In low-reciprocity groups, self-confident workers were more likely than others

[37] Both of these variables were measured by sociometric choices: workers named by others as consultants and workers named by others as asking questions.

[38] The index of self-confidence was whether or not a worker stated he could do his job without help from the supervisor.

TABLE 12. RECIPROCITY IN CONSULTATIONS, SELF-CONFIDENCE, AND PER CENT REGULARLY CONSULTED

Worker Self-Confident	*Reciprocity in Group*	
	High	*Low*
Yes	41% (17)	73% (15)
No	77% (13)	47% (15)

to be regularly consulted by colleagues, just as highly competent workers were. This association again is not unexpected. The more competent caseworkers tended to be more self-confident, and a self-assured bearing as well as technical skills increased the chances that a caseworker would impress others as an expert and, consequently, become a popular consultant. Besides, being regularly consulted by colleagues probably also raises a worker's self-confidence. What is unexpected, however, is that this same association was not present in high-reciprocity groups. On the contrary, there self-confidence was inversely associated with being much consulted by colleagues. We have suggested in the preceding section that the ego-threat posed by frequently asking colleagues for help is what motivates workers to establish consultation partnerships, and this interpretation can also account for the finding here. Workers whose self-confidence had been undermined had most incentive to develop reciprocal consulting arrangements, and they probably found other modest workers more attractive partners than self-assured workers who would be threatening to them. Hence, self-confidence decreased the likelihood of being regularly consulted in groups where reciprocal partnerships prevailed, although it increased this likelihood in groups where most consultations were unilateral. Moreover, being regularly consulted in a reciprocal partnership, as distinguished from being often asked for advice in a non-reciprocal situation, is not particularly apt to promote self-confidence.[39]

Homophily and Heterophily in Consultation. An earlier chapter stressed the importance, for operations, of the orientations of members of the organization to their work. Now we shall briefly consider the significance of these orientations for the effectiveness of consultations among colleagues. The term "homophily" has been used to refer to friendships between persons having similar values or other characteristics.[40] We shall use the concepts of homophily and heterophily to refer to the similarity or difference in value orientations between associates.

39 How typical these patterns are is, however, questionable, since data from County Agency did not replicate them.

40 See Paul F. Lazarsfeld and Robert K. Merton, "Friendship as Social Process," Morroe Berger *et al.* (eds.), *Freedom and Control in Modern Society,* New York: Van Nostrand, 1954, p. 23.

The question we want to raise is: which benefits performance most, consulting colleagues who have values similar to one's own, or consulting those with different values? Pelz's study of a government medical-research organization attempts to answer this question.[41] The researchers were classified as being either "scientific" or "institutional" in their orientation. In general, these two orientations bear a close resemblance to Gouldner's "cosmopolitan" and "local" orientations, respectively.[42] Pelz averaged the orientations of the five colleagues named by each scientist as the ones with whom he often discussed his work, and also computed the mean frequency of contacts each scientist reported with those colleagues. It was found that heterophily of orientations among consultants improved scientific performance. Specifically, if the average orientation of a researcher's five colleagues was similar to his own, then frequency of consultation with them was not related to his performance, but if their orientations were different from his, then frequent contacts with them improved his performance. The results were reversed, however, if only the orientation of the one closest colleague was considered. If the closest colleague's orientation was different from that of the respondent, frequency of contact was inversely associated with performance—discussions seemed to be a waste of time. But if the two individuals shared the same orientation, then frequency of contacts was directly associated with research performance.

Pelz interprets these findings as indicating that research performance depends upon the challenging stimulation as well as the support the social environment affords. Discussions with one's closest colleague should furnish social support to facilitate thinking and reduce anxieties. They appear to do this if the colleague's orientation is similar to one's own. If his orientation is different, he is likely to play the role of critic, making discussions with him disturbing so that frequent contacts with him will impede performance. Discussions with other less closely related colleagues, on the other hand, should supply challenges and serve to correct errors resulting from the one-sided viewpoint produced by a person's own orientation. A person's social contacts with colleagues whose orientation is the same as his own will fail to accomplish this, whereas frequent discussions with heterophilous associates in the organization will furnish such stimulation and thus improve research performance. Other data presented support this interpretation. For example, if the field of specialization of a scientist's closest colleague and that of his superior were different from one another, his performance tended to be better than if they were similar

[41] Donald C. Pelz, "Some Social Factors Related to Performance in a Research Organization," *Administrative Science Quarterly*, 1 (1956), pp. 310–325.

[42] See above, at pp. 64 and 66, especially footnotes 6 and 12, Chapter III.

to one another (whether both were similar to or different from his own specialty). Thus, optimum performance resulted if one of the two people with whom a person consulted most often was similar and the other was dissimilar in orientation to him. One provides social support; the other provides challenges and an opportunity for correction of errors.

Suggestive as these conclusions are, they remain only plausible hypotheses at this point. They rest on the assumption that performance is affected by homophily or heterophily, but the direction of influence may be the reverse. For instance, the most competent scientists, those who perform best, may be less reluctant to consult colleagues whose orientations differ from theirs than are researchers who are less competent and consequently less secure. Although the combination of findings reported increases the plausibility of Pelz's interpretations, his survey data cannot demonstrate the validity of his causal inference.

Concluding Remarks

The central topic of this chapter was the significance of the free flow of communication among colleagues. By providing social support, challenging stimulation, error correction, and a *laissez-faire* competition for respect among participants, the free flow of communication contributes to finding solutions to problems, to making decisions, and to creative thinking. But the battleground of ideas generated by such a free flow makes coordination more difficult. Hierarchical differentiation of status tends to block the free flow of communication in a group and thus to impede problem-solving, although some types of leadership have the opposite consequences, such as that of the discussion chairman who prevents suppression of minority opinions.

A hierarchical organization, in part precisely because it restricts the free flow of communication, improves coordination; indeed, it seems to be essential for effective coordination of group effort. This is the dilemma posed by hierarchical differentiation: while it is necessary for coordination, it blocks the communication processes that are vital for stimulating initiative and facilitating decision-making. Moreover, even if there were no formal hierarchy in the organization, communication among peers would be likely to give rise to informal differentiation of status, which also creates obstacles to communication. These considerations illustrate the dynamics of organizational life. Hierarchical structures channel communication processes; communication processes give rise to structural differentiation, and the emergent social structures, in turn, redirect the communication processes. The existence of dilemmas that cannot be definitively and finally resolved is inherent in such dynamic processes.

CHAPTER SIX

The Role of the Supervisor

The authority of superiors in a formal organization is legitimated by legal contract rather than by traditional values or by an ideological identification with a charismatic leader. Employees assume the contractual obligation to follow managerial directives. Commons, in his famous discussion of the wage bargain, states that what the worker sells "when he sells his labor is his *willingness* to use his faculties according to a purpose that has been pointed out to him. He sells his promise to obey commands."[1] The directives controlling a subordinate include the various administrative regulations, the professional or technical principles guiding his activities, and, particularly, they include the orders he receives from his superiors.

The formal authority that has its source in this legal contract, however, is extremely limited. First, as Commons points out,[2] the labor contract is not really an enduring commitment of the employee, since he is free to leave his job if he can find a better one, and neither is it such a commitment on the part of the employer unless tenure provisions or union agreements specifically prohibit dismissals at his free discretion. Second, by its very legal nature, the contract obligates employees to perform only a set of duties in accordance with minimum standards and does not assure their striving to achieve optimum performance. The legal authority of management to assign tasks to employees is rarely questioned—there is willing compliance—but this legal authority does not and cannot command the employee's willingness to devote his ingenuity and energy to performing his tasks to the best of his ability. Important as formal authority is for meeting the minimum requirements of operations in a complex organization, it is not sufficient for attaining efficiency. It promotes compliance with directives and discipline, but does not encourage employees to exert effort, to accept responsibilities, or to exercise initiative.

The narrow scope of formal authority induces management, in the interest of effective operations as well as in its self-interest, to seek to widen the sphere of its influence over employees beyond the con-

[1] John R. Commons, *Legal Foundations of Capitalism,* New York: Macmillan, 1924, p. 284 (italics in original).
[2] *Ibid.,* pp. 284–286.

trolling power that rests on the legal contract or on formal sanctions. For this reason many books on management place heavy emphasis on administrative leadership rather than sheer legal authority.[3] They stress the importance of such executive functions as defining the goals and responsibilities of the members of the organization, inspiring them to identify with the objectives of the enterprise and to pursue them to the best of their abilities, motivating them to collaborate for this purpose, and resolving conflicts that may arise in the organization. Executive leadership evidently involves exerting influences that go far beyond the confines of the legal contract and thus cannot be legitimated by it. But, as a recent analysis by Bendix shows, managerial ideologies develop in modern societies that serve to justify the power of management in terms of basic cultural values.[4] These value orientations legitimate administrative leadership and, thereby, transform it into managerial authority.

The problem of finding ways to extend the scope of formal authority is not one that confronts higher management only: it also is a challenge that faces first-line supervisors. For the supervisor cannot effectively discharge his responsibilities without exerting more influence on his subordinates than his formal authority alone permits. The "style of leadership" of a supervisor is of great importance, since it governs the amount of influence he has over subordinates in addition to the power that derives directly from his formal status in the organizational hierarchy. Indeed, these variations in effective authority are what make the empirical study of supervision an interesting and fruitful task for the sociologist. If there were no such variations resulting from differences in leadership styles and if official authority alone determined supervisory practices, everything about them could be learned simply by reading the official procedure manuals and job descriptions.

Styles of Supervision

Formal Authority and Leadership. A problem facing all supervisors is to find ways of extending the scope of their influence over subordinates beyond the narrow limits of formal authority. One possible strategy is that of domination. A supervisor may extend his controlling power over subordinates beyond the willing compliance his legal status com-

3 For a classic and a more recent work stressing managerial leadership see Chester I. Barnard, *The Functions of the Executive,* Cambridge, Mass.: Harvard University Press, 1938; and Philip Selznick, *Leadership in Administration,* Evanston, Ill.: Row, Peterson, 1957.

4 Reinhard Bendix, *Work and Authority in Industry,* New York: Wiley, 1956.

mands by resorting to formal sanctions, or to threats of using these sanctions. Quite a different strategy is that of leadership, in which a superior furnishes services that obligate subordinates to him. A supervisor has many opportunities for assisting subordinates. He is responsible for training and advising them, and they will appreciate it if he exerts special effort in doing so and is easily accessible when they need help. He can facilitate their work, for example, by getting tools in short supply or quickly locating a repairman when a machine breaks down. He may back his own work group in interdepartmental conflicts and represent them and their cause to the administration, and he may perform special favors, such as adjusting the vacation schedule to fit the personal preferences of workers. Rendering these kinds of services creates social obligations, and subordinates who are obligated to their supervisor for benefits received will feel they should reciprocate by complying with his requests and special demands. In this manner, the effective leader increases the sphere of his influence over subordinates.

The formal status and official powers of the supervisor help him to provide services to subordinates. To cite an obvious example, his formal status in the organization, which gives him greater access to management and other supervisors than his subordinates have, enables the supervisor to channel needed information, supplies, and services from other departments to his subordinates. But the supervisor who resorts to formal status prerogatives and sanctions in his relations with his subordinates will alienate them. He can use his formal powers to coerce them to do his bidding, to be sure, but this method of supervision is hardly designed to create social obligations and promote a willing compliance with his requests. Far from strengthening his authority, such authoritarian behavior undermines it. Indirectly, however, the supervisor can use the official powers granted him so as to broaden his influence over subordinates and promote willing compliance with his directives, for he can create social obligations by judiciously *refraining* from using all his power of control. For example, a supervisor who does not enforce an unpopular no-smoking rule obligates subordinates to him; but if this rule had not existed, or if he had not had the power to enforce it, he could not have established this social obligation. In short, one function of formal rules and status prerogatives in organizations, although clearly not the only one, is that they bestow powers upon the supervisor that enable him to obligate subordinates and win their good will simply by refraining from using these powers.

Merton points out that effective and stable authority requires that superiors be in a position to observe the actual performance of sub-

ordinates and become familiar with their prevailing norms.[5] But he adds that this visibility must not be unrestricted lest it create opposition which impedes effective operations, because subordinates demand some leeway to depart from the ideal requirements of their office. This proviso implies that any departure from official rules that becomes visible to the superior will be suppressed by him; however, it is not necessary for supervisors to act in this fashion. Our analysis suggests that it may not be visibility that must be limited but rather the superior's tendency strictly to enforce all official rules. The greater the field of vision of the superior—his knowledge of formal requirements, group norms, and actual practice—the greater will be his ability to judge which operating rules can be ignored without impairing efficiency and which unofficial practices are rooted in such basic work-group norms that he cannot afford not to tolerate them. Thus, the supervisor's greater range of vision enables him to decide rationally in terms of his responsibility which official rules he must enforce and which he can permit to be violated by subordinates in order to establish social obligations.

Influence over subordinates acquired by obligating them does not constitute an established authority over them, although it may lead to this. For authority requires legitimation. And while authority can be exercised in pair relations, it can originate only in a group because only a group can provide legitimation of the control exercised. What social process legitimates the control exercised by a supervisor? The supervisor's ability to help subordinates solve complex problems commands their respect; and his willingness to furnish help and do favors for them commands their allegiance. As the members of the work group share their respect for and loyalty to the supervisor, there develops a consensus among them that they should comply with his wishes and suggestions. Once established, these norms of allegiance and respect are enforced by the group because all might suffer if some members failed to repay their obligations to the supervisor. In this fashion, compliance within certain bounds becomes a group norm—internalized by group members and enforced by group sanctions. These values legitimate the extension of the supervisor's authority beyond the legally prescribed limits.[6]

Formal and informal authority can now be clearly distinguished. Authority has been earlier defined as the exercise of control that rests on the willing compliance of subordinates with the directives of their superior. To say that this compliance is voluntary, however, does not

5 Robert K. Merton, *Social Theory and Social Structure* (2d ed.), Glencoe, Ill.: Free Press, 1957, pp. 339–350.

6 A similar point is made by George C. Homans, in his *Social Behavior: Its Elementary Forms,* New York: Harcourt, Brace, 1961, pp. 294–295.

imply that it is independent of social constraints. On the contrary, the social value orientation that legitimates the exercise of authority constrains subordinates to obey the superior's orders. They voluntarily follow the mandates of their own values, which have become internalized, and which are enforced through social sanctions. Formal authority is legitimated by values that have become institutionalized in legal contracts and cultural ideologies, and the social constraints that demand compliance pervade the entire society. Informal authority, on the other hand, is legitimated by the common values that emerge in a group, particularly by the loyalty the superior commands among group members, and group norms and sanctions enforce compliance.

This theoretical analysis implies that superiors who command the loyalty of their subordinates are more likely than others to establish effective informal authority over them and thus to influence them. A number of empirical findings support this conclusion. For example, an experiment conducted by French and Snyder, which used groups composed of one supervisor and three members engaged in a group judgment test, indicates that the more accepted the leader was by the rest of the group, the more he attempted to influence them and the more successful his attempts to influence were.[7] In addition, group respect for the ability (intelligence) of the leader was associated with successful influence attempts. Similar results were obtained in a field experiment conducted by Lippitt and his colleagues.[8] Boys in a camp setting to whom other boys attributed much power made more influence attempts and in addition were more successful in these attempts. Further, the boys with high power were liked better than other group members and were more frequently identified with.

The implication of these findings is that superiors who command the loyalty of a group—who are liked, accepted, respected—have more control than others over group members partly because they have greater confidence in their authority to issue directives—influence attempts were more frequent as well as more successful. Since the ultimate criterion of effective supervisory authority is the performance of subordinates, our interpretation also implies that superiors who command loyalty will have more productive groups than those who do not. This interpretation was found to apply in the French and Snyder experiment: on a task involving card sorting, the acceptance of the leader was positively correlated with the productivity of the group ($r =$

[7] John R. P. French, Jr., and Richard Snyder, "Leadership and Interpersonal Power," Dorwin Cartwright (ed.), *Studies in Social Power*, Ann Arbor: Institute for Social Research, University of Michigan, 1959, pp. 118–149.

[8] Ronald Lippitt *et al.*, "The Dynamics of Power," *Human Relations*, 5 (1952), pp. 37–64.

+.42).[9] In City Agency, work groups whose supervisor commanded the loyalty of his subordinates performed better in various respects than work groups with a supervisor who failed to inspire loyalty. For example, groups with high loyalty to the supervisor were more productive than those with low loyalty, as indicated by both the number of field visits per month and the proportion of time spent in the field.

Measuring Differences in Supervision. Various research techniques can be used to study differences in supervisor practices in a formal organization. Three methods of major interest are: (1) the direct observation of the supervisor's interaction with his subordinates; (2) interviews with supervisors; and (3) interviews with those in contact with the supervisor—his subordinates, his superiors, or his co-workers. We shall consider each of these methods briefly.

Several studies have employed direct and indirect observation of supervisors as they go about their daily work. Barnard, for example, relies for most of his empirical material on his recollections of processes of management and supervision as a participant.[10] Lasswell, on the other hand, has utilized a method whereby supervisors record their own interaction in a systematic fashion—the contacts they have with others, their length, and their significance—as they carry out their responsibilities.[11] Barnard's method of participant observation is suitable for exploratory purposes—gaining new insights and suggesting hypotheses—but not for testing hypotheses, since it does not yield systematic data. Lasswell's approach may provide systematic data but has the shortcoming that it requires participants continually to change the focus of their attention from being a supervisor to becoming an observer and back again. The frame of reference needed for successful performance of one role would appear to conflict with effectively carrying out the other. As a result, the quantitative data may be of questionable reliability and validity. This specific problem can be overcome by introducing an outside investigator to observe the process of supervision in organizations, as Whyte did in his study of restaurants.[12]

Only quite recently have students of industry and administration begun to systematize the observation of supervision by adapting the quantitative methods of observing interaction in peer groups that have been developed in experimental laboratories to the study of supervisory processes in field situations. Thus, Lawrence has modified the Bales category system of interaction-process analysis for use in the

[9] French and Snyder, *op. cit.*, p. 146.

[10] Barnard, *op. cit.*

[11] Harold D. Lasswell, "Self-Observation," Robert K. Merton *et al.* (eds.), *Reader in Bureaucracy,* Glencoe, Ill.: Free Press, 1952, pp. 425–430.

[12] William Foote Whyte, *Human Relations in the Restaurant Industry,* New York: McGraw-Hill, 1948, pp. 359–368.

investigation of interaction between supervisors and subordinates.[13] Carter and his colleagues use a large number of categories divided into seven principal dimensions for the study of leadership in face-to-face groups.[14] And Wirdenius has developed and tested in field settings a method of classifying supervisory behavior.[15] Chapple's method for measuring interaction has received fairly widespread use in field situations because of its simplicity.[16] His system provides data on the frequency of interactions between various participants and the direction of interaction (who initiates each contact); the data yielded by the resulting who-to-whom matrix can be used to analyze, among other things, the role of the supervisor in his own work group and his relations to his peers and superiors. Such a quantitative record of the patterns of interaction that occurred in the office was kept for 12 of the 23 work groups in the two welfare agencies.[17] In addition, unit meetings and conferences between supervisors and workers were observed in 13 of the work groups, and quantitative measures on these sessions are available for 5 of these groups. In general, these data were used to validate other measures of supervision which are available for all (or nearly all) supervisors.

Data on supervision can also be obtained through interviews with supervisors; their attitudes, orientations, and descriptions of their role performance may be elicited in this manner. Interviews, like observation, may provide more or less systematic data. The studies conducted by the Survey Research Center provide a good illustration of an attempt to collect systematic data on supervisory practices by means of interviews with supervisors.[18] An important limitation of this method, particularly if it is used as the exclusive source of information on supervision, is that it provides a one-sided view of a social relationship,

[13] Paul Lawrence, *The Changing of Organizational Behavior Patterns,* Boston: Graduate School of Business Administration, Harvard University, 1958, pp. 130–141. Caudill has also utilized the Bales categories in the study of interaction in a field setting. See William Caudill, *The Psychiatric Hospital as a Small Society,* Cambridge, Mass.: Harvard University Press, 1958, pp. 231–265.

[14] Launor Carter *et al.,* "The Relation of Categorizations and Ratings in the Observation of Group Behavior," *Human Relations,* 4 (1951), pp. 239–254; and Carter *et al.,* "A Note on a New Technique of Interaction Recording," *Journal of Abnormal and Social Psychology,* 46 (1951), pp. 258–260.

[15] Hans Wirdenius, *Supervisors at Work,* Stockholm: The Swedish Council for Personnel Administration, 1958, pp. 86–111.

[16] Eliot D. Chapple, "Measuring Human Relations," *Genetic Psychology Monographs,* 22 (1940), pp. 3–147.

[17] The formal observation period was approximately 25 net hours in each agency.

[18] Robert L. Kahn and Daniel Katz, "Leadership Practices in Relation to Productivity and Morale," Dorwin Cartwright and Alvin Zander (eds.), *Group Dynamics,* Evanston, Ill.: Row, Peterson, 1953, pp. 612–628.

presenting only the perspective of the superordinate participant. Besides, the supervisor's perceptions of his activities and relations with others is subject to distortion by his attitudes and orientations. As a matter of fact, the artificial interview situation may not even result in accurate information about the supervisor's orientation, since he may continue, so to speak, a controversy he has had with colleagues in the past as he responds to a salient question. When asked whether supervisors should be businesslike or informal in their relations with workers, for example, some of those we interviewed began by saying, "Of course, it is important to be businesslike . . ." Only after they had considerably elaborated this point and their attitudes had been further probed did it become apparent that this introductory statement was a kind of strategic ploy utilized to ward off criticism that had presumably been previously directed at them for favoring a very informal approach. They were, in effect, saying, "I do too recognize that being businesslike sometimes is important." In a short-answer interview, the actual preference for an informal approach underlying such statements might remain undiscovered.

An extension of the principle underlying Moreno's sociometric method[19] is to ascertain a supervisor's approach and practices by asking his subordinates about them.[20] This method was used in our studies of the two welfare agencies. To illustrate this procedure: subordinates are asked whether they consider their supervisor to be easy or strict, and supervisors are then classified on the basis of whether the majority of their subordinates select the one or the other response. Of course, it is not known whether or not all supervisors classified as strict by this criterion would be found to be more strict on the basis of some objective measure than those categorized as easy. But this method does furnish a measure of social reality—it does tell us which supervisors are socially defined as being strict by their subordinates. (Instead of using the majority opinion, the median can be used if necessary to obtain a better distribution of supervisors.)

More complex methods of assessing the role of supervisors than those discussed have been utilized. For example, Foa has developed a measure of the supervisor's role which combines the responses of both the superior and his subordinates.[21] One of the measures developed in our research, an index of hierarchical independence to be described

[19] Jacob L. Moreno, *Who Shall Survive?* Washington, D.C.: Nervous and Mental Disease Publishing Co., 1934.

[20] This approach is utilized in A. L. Comrey *et al., Factors Influencing Organizational Effectiveness,* Los Angeles: University of Southern California, Final Technical Report, The Office of Naval Research, 1954, pp. 12–41.

[21] Uriel G. Foa, "The Foreman-Worker Interaction: A Research Design," *Sociometry,* 18 (1955), pp. 226–244.

later, also combines data from two sources to characterize supervisors.

Contrasting Styles of Supervision. The study carried out in City Agency included the analysis of several aspects of supervision, systematic data being collected on all twelve supervisors in two general-assistance field sections. A considerable number, though not all, of these supervisory characteristics were also examined in the study of eleven supervisors in County Agency; again only supervisors in the field sections were included. The concern of our analysis is with the interrelations between the various supervisory characteristics and with the effects of differences between supervisors on the groups of subordinates.

Data from City Agency suggest that a number of supervisory characteristics were so closely interrelated that they seem to define an overall syndrome of supervisor authoritarianism: supervisors who displayed these characteristics were labeled "authoritarian"; those who lacked them, "nonauthoritarian." An authoritarian approach, which is not to be confused with the exercise of effective authority, involves a somewhat rigid and domineering pattern of close supervision. The distinction was based on the responses of subordinates when asked to characterize their own supervisor.[22] Seven items were found to be closely related; together they constitute the operational definition of authoritarianism.[23] The authoritarian supervisor was considered to be strict rather than easy by most of his subordinates; he was felt to supervise closely rather than to let subordinates work pretty much on their own; the majority of workers did not describe his approach to them as friendly or in favorable terms; his instructions were felt to be sometimes unclear; he was said to stick closely to procedures rather than be willing to make reasonable exceptions to rules; his subordinates did not consider his approach to clients to indicate an orientation to casework; and, finally, some of his subordinates explicitly stated that he was authoritarian in his approach.

Validation of this syndrome of authoritarianism was attempted in several ways. Four of the above items were included in the County Agency study, and the same close interrelations between them were found there. Moreover, a number of observational measures were re-

[22] Responses on a large variety of supervisory practices were elicited in the interview, some in open-ended questions, others in paired comparisons (the workers being asked to choose the one of two descriptive statements that best characterized their supervisor), and still others in a list of statements the workers were asked to check if applicable.

[23] All items were dichotomized and relationships between any two were established in two-by-two contingency tables. Two items were considered to be related if ten or more of the twelve supervisors fell into a diagonal. Each of the seven items included in the syndrome was related to at least four others. No supervisor classified as authoritarian had more than two (out of a possible seven) nonauthoritarian characteristics, and vice versa.

lated as expected to this index of authoritarianism. An authoritarian supervisor was more likely than another to take the initiative in interaction with his subordinates, as revealed by the proportion of his contacts with workers originated by him rather than by them. This tendency probably indicates that authoritarian supervisors were more apt to check on workers instead of waiting for their subordinates to come to them with their problems. A quantitative record showed that caseworkers under authoritarian supervisors were more prone than other workers to report to the supervisor immediately after returning to the office from the room where they interviewed clients. Apparently these workers found it necessary to check with their supervisor before taking action and did not feel as free as workers under less authoritarian supervisors to act on their own initiative. There was a further indication of the unwillingness of subordinates under authoritarian supervisors to assume responsibility; observation of worker-supervisor conferences revealed that workers under authoritarian supervisors disproportionately often consulted the supervisor even when they already had the answer to their problem. Finally, a record of the discussions at work-group meetings indicated that authoritarian supervisors were more likely than others to issue orders to their subordinates: a larger proportion of their statements in group meetings were directives. Their style of supervision was to tell workers what to do rather than to use the meetings for explaining issues and letting the group explore them or for discussions of problems raised by subordinates.

Two other important role attributes of supervisors are their social distances from subordinates and superiors, respectively. A supervisor's approach to his subordinates was defined as "detached" if a majority of them said that he "always remains calm and never really loses his temper" rather than that "he can get quite excited." Detachment seems to have been a salient and clearly visible role characteristic of supervisors in City Agency, as shown by the very high consensus among subordinates: seven supervisors were looked upon by subordinates as detached and five as excitable; only five of sixty caseworkers endorsed statements that disagreed with those of the majority in their unit.[24]

The concept of hierarchical independence refers to the fact that a supervisor maintains in his work a minimum degree of independence from his superior, the section chief. In order to determine the supervisor's independence, five of the characteristics of his role performance as defined by his subordinates were compared with the same five characteristics of his section chief as defined by the chief's subordinates (the

24 There was a slight inverse relationship between detachment and authoritarianism, but it did not meet the criterion established for inclusion in the authoritarian syndrome (see footnote 23). In County Agency, few respondents said their supervisor was excitable (14 per cent), and consensus on this item was not as high as in City Agency.

supervisors directly under him). The five characteristics involved were: sticking closely to procedures rather than making reasonable exceptions; supervising closely rather than letting subordinates work on their own; a formal as opposed to an informal approach to subordinates; extensive knowledge of procedures as opposed to a lesser concern with knowing rules; and being strict as opposed to being lenient. Since the two section chiefs differed greatly with regard to these characteristics, the index of hierarchical independence reveals nothing about the style of leadership of different supervisors but measures specifically the amount of difference between a supervisor's approach and that of his superior.[25]

We shall examine the significance of different supervisory role characteristics by relating them to several effect criteria, using the findings from studies of various organizations reported in the literature as well as those from our welfare-agency research. The main effect criteria to be considered are: the loyalty the supervisor commands among his subordinates; the productivity in his unit; other aspects of the performance of his subordinates and their approach to their work, such as whether they are oriented toward service or toward procedure; the work satisfaction of his subordinates; and the group solidarity that prevails in his unit.

Supervision and Performance

Effects of Authoritarian Practices. Recent studies carried out in organizational settings indicate that authoritarian styles of supervision lessen worker satisfaction and productivity. Studies in an insurance company and a tractor-manufacturing firm conducted by the Survey Research Center, for example, found that closeness of supervision was inversely related to productivity.[26] Group productivity was negatively affected not only if workers were closely supervised by their foremen, but also if foremen were closely supervised by their superiors; apparently pressures were transmitted from level to level. In the tractor company, closeness of supervision was also inversely related to the amount of job satisfaction reported by workers. An experimental study conducted within a telephone company by Westerlund reports that workers allowed to work under informal supervision in small groups showed an increase in productivity over those operating under more rigid, departmental supervision.[27]

[25] The criterion of independence is that the supervisor's role differed from that of his section chief on two or more of the five items. The data necessary to construct this index were not collected in County Agency.

[26] Kahn and Katz, *op. cit.*, pp. 617–620.

[27] Gunnar Westerlund, *Group Leadership,* Stockholm: Nordisk Rotogravyr, 1952.

Comrey and his associates conducted twenty-nine questionnaire studies of different types of organizations. Summary results are based on the eleven studies that showed the highest proportion of significant findings.[28] The results pertaining to authoritarianism were that the most effective supervisors differed from others, according to the reports of subordinates, in the following respects: they communicated more information to their subordinates, were considered to be helpful, were not arbitrary in their decisions and judgments, and had a sympathetic approach to workers. Finally, a study by Argyle and colleagues of ninety roughly comparable work groups concludes on the basis of the empirical evidence that the combination of general, democratic, and nonpunitive qualities in supervision was most conducive to high productivity in a work group.[29] However, as a result of his own studies, and after a survey of the literature, Argyle cautions that the differences in productivity in work groups resulting from contrasting methods of supervision were typically small, usually not larger than 15 per cent of the total output.

In the two welfare agencies, authoritarian supervision appeared to decrease work satisfaction, as indicated by worker expectations to leave the agency. Thus, in City Agency 65 per cent of 29 workers under authoritarian supervisors, as compared with 48 per cent of 31 workers under nonauthoritarian supervisors, expected to leave the agency in less than ten years. In County Agency, differences were smaller but in the same direction: 31 per cent of the 52 workers under authoritarian supervisors and 23 per cent of the 27 workers under nonauthoritarian supervisors expected to leave the agency in less than five years.[30]

Worker orientation to clients was also affected by authoritarian supervision. Authoritarian supervisors, who were invariably the procedure-oriented ones,[31] seem to have encouraged a concern among their subordinates with checking the eligibility of clients at the expense of rendering casework services. In City Agency, only 24 per cent of the 29 workers under authoritarian supervisors, in contrast to 61 per cent of

[28] Comrey *et al., op. cit.,* p. 55.

[29] Michael Argyle *et al.,* "Supervisory Methods Related to Productivity, Absenteeism, and Labour Turnover," *Human Relations,* 11 (1958), pp. 23–40.

[30] The proportion of all workers expecting to leave differed in the two agencies undoubtedly in part because the wording of the question was different (ten years as opposed to five). However, these proportions probably also reflect differences in the availability of outside employment opportunities in the two communities (see above, p. 114), as well as actual differences in worker satisfaction, which appeared to be higher in County Agency.

[31] Differentiation between supervisors on the seven-item index of authoritarianism was found to be perfectly correlated with their differentiation on one of the seven items: all authoritarian supervisors, and none of the others, were thought by most of their subordinates to "stick closely to procedures."

the 31 caseworkers under the nonauthoritarian supervisors, were oriented to casework service.[32] This particular information was not gathered for County workers, but a different indicator of orientation to clients was available for both agencies. Visits with nonrecipients (collaterals) are usually conducted for the purpose of verifying information obtained from the client or for checking up on him in other ways.[33] In both agencies, workers under authoritarian supervisors made more visits to collaterals per month than did workers under nonauthoritarian supervisors. In other words, authoritarian methods of supervision tended to foster a narrow concern with enforcing eligibility procedures and to discourage a willingness to assume a broader responsibility for service. This conclusion is further supported by the finding that authoritarianism curtailed the desire of subordinates to assume more responsibility than they presently had. In City Agency, 52 per cent of the workers under authoritarian supervisors but 75 per cent of those under nonauthoritarian ones indicated an interest in extending the scope of their responsibilities. In County Agency, the corresponding proportions of workers who wanted increased responsibility were 36 and 45 per cent, respectively.[34]

Contrary to expectations, work-group productivity as measured by

[32] This type of relation raises an important methodological question. The finding that a supervisory characteristic *as defined by responses of subordinates* is related to the orientation of these subordinates may not indicate, as here assumed, that supervisory practices influence the orientation of workers but rather that these orientations find expression in the worker's perception of his supervisor. The specific relation examined, for example, may not be due to the influence of authoritarian supervision on the worker's orientation to clients but to the fact that workers who themselves are more rigid in their approach are more likely than other workers to describe their supervisor as procedure-oriented and as having other authoritarian characteristics. But when we statistically control the way in which a given worker describes his supervisor, we still find that the worker is less apt to be casework-oriented if he works under a supervisor whom other workers define as authoritarian than if he works under a nonauthoritarian supervisor. This finding indicates that the actual role of the supervisor, and not the worker's subjective perception of this role, is responsible for the relationship observed. Only relations for which this has been found true are reported.

[33] This statement does not apply for workers with Old Age Assistance caseloads in County Agency. Hence, OAA workers were excluded from this analysis.

[34] Differences in operating procedures in the two agencies made it necessary to ask different questions to ascertain the worker's interest in assuming more responsibility. In City Agency, workers had to submit the first assistance budget they worked out for a new client to the supervisor for his signature. Here the question asked was, "If they would permit you to sign your own first budgets, would you want to?" In the County Agency, so-called "independent" workers were permitted to make more decisions on their own than others, and the question asked was, "Would you like to be an independent worker?"

the numbers of visits to recipients per month, was not related to authoritarianism. Only small differences were found in the two agencies, and they were in opposite directions. This finding conflicts with the findings of other studies. Perhaps in a service organization, in contrast to factories and private offices, employees do not react to authoritarian supervision by reducing the effort they devote to their work. Neither was authoritarianism negatively related to commanding the loyalty of subordinates (as indicated by their preference for their present supervisor), another relationship which we had expected to observe. While this conclusion is based on data from City Agency alone,[35] data from both agencies are available on whether subordinates considered their supervisors helpful. In neither agency was there any association between the supervisor's authoritarianism and the tendency of his subordinates to describe him as helpful. In short, although authoritarian practices had an adverse effect on the orientations of subordinates to their job, their work, and their clients, these practices did not alienate subordinates as much as had been expected.

Effects of Detachment and Independence. The human-relations approach has correctly emphasized the importance of social relations for operations. We have, for example, just examined the dysfunctions of authoritarian supervisory practices for the tendency of subordinates to render service, assume responsibilities, and experience work satisfaction. But this approach has tended to ignore other important requirements of effective supervision. Maintaining "good" human relations is not the only prerequisite for being an effective manager of men. Indeed, there is a danger that relations between supervisors and subordinates will be "too good," since the supervisor may become personally involved in the network of relations among subordinates to the extent of being confined by them. Thus, the first manager in the gypsum firm studied by Gouldner had many informal contacts with workers and was on very good terms with them, but his consequent indulgent methods constituted ineffective management.[36] He seems to have succumbed to the temptation of letting a concern for maintaining good human relations interfere with his managerial responsibility for the operating efficiency in the plant. Effective leadership requires not only refraining from closely checking on subordinates but it also requires making challenging demands to stimulate their interest and, indeed, their ability to perform well.

These considerations imply that some social distance and independence from subordinates promotes effective leadership. This is the

[35] The data on loyalty to the supervisor were not collected in County Agency.

[36] See Alvin W. Gouldner, *Patterns of Industrial Bureaucacy,* Glencoe, Ill.: Free Press, 1954, pp. 45–56.

conclusion Fiedler draws from a series of studies.[37] Various types of leaders were given a test in which they selected words that characterized the co-workers they most and least preferred. Low scores on this test of Assumed Similarity between Opposites (resulting from large differences in the descriptions given) have been interpreted as a generalized index of social distance between the subject and the rest of the group, since they indicate the subject's tendency to discriminate between group members rather than to approve of all of them. Such different groups as basketball teams and surveying parties were more successful if their informal leader had a low Assumed Similarity score than if he had a high score. The same results were obtained for open-hearth shop foremen. Fiedler concludes that "leaders who perceive themselves to be more distant from their subordinates have more effective work units than those who perceive themselves to be closer to their men."[38]

Another indicator of a lack of involvement with subordinates is the supervisor's ability to maintain emotional detachment—to remain calm and rarely, if ever, lose his temper. Such detachment was perfectly associated with commanding loyalty in City Agency. All seven detached supervisors, but none of the five excitable ones, commanded the loyalty of most of their subordinates, that is, were preferred by them over any other supervisor, and were also most respected by them. The index of helpfulness, available for both agencies, confirms the conclusion that subordinates appreciated detachment in their supervisors. In County Agency, 85 per cent of the 55 workers under detached supervisors as opposed to 47 per cent of the 19 workers under excitable ones, considered their supervisor to be helpful. In the City, 56 per cent of 36 workers under detached supervisors and 17 per cent of 24 workers under excitable ones felt that their supervisor was helpful.

Detachment, unlike authoritarianism, was related to productivity

[37] Fred E. Fiedler, "A Note on Leadership Theory," *Sociometry*, 20 (1957), pp. 87–94.

[38] *Ibid.*, pp. 92–93. These are important insights but they are not as yet supported by other research. Comrey and his colleagues, for example, found no relationship between social distance and effectiveness of supervision (*op. cit.*, p. 40). Fiedler's proposition that the *superior's* perception of great social distance promotes *actual* performance is also somewhat at variance with, although not directly contradicted by, the finding of studies by the Survey Research Center that *subordinates* who perceive their superior as not distant have more favorable *attitudes* to the job than others (see Rensis Likert, "A Motivational Approach to a Modified Theory of Organization and Management," Mason Haire (ed.), *Modern Organization Theory*, New York: Wiley, 1959, p. 189). Finally, other studies by Fiedler himself dealing with bomber crews, tank crews, and farm-supply cooperatives suggest that optimum group performance results from certain combinations of sociometric closeness and psychological distance between leaders and "keymen" (group members whose behavior most directly influences team performance). See Fred E. Fiedler, *Leader Attitudes and Group Effectiveness*, Urbana: University of Illinois Press, 1958, pp. 27–45.

in both agencies (see Table 13). Two-thirds of the workers under detached supervisors, but only about one-third of the workers under excitable supervisors exerted much effort in making field visits to recipients. Worker satisfaction, however, was hardly associated with the supervisor's detachment in City Agency and not at all in County Agency.

TABLE 13. WORKER VISITS TO RECIPIENTS AND ORIENTATION OF SUPERVISOR

Orientation of Supervisor	*Workers with Many Recipient Visits*		
	City Agency	*County Agency*	*Total*
Detached	69% (28)	66% (38)	67% (66)
Excitable	29% (17)	44% (16)	36% (33)

Supervisors, like foremen in industry, are men in the middle, subject to pressures from above as well as below, and their ability to exercise their leadership functions requires some independence from both kinds of pressure. Maintaining a degree of independence from superiors is no less important for effective supervision than remaining somewhat detached toward subordinates. Thus, Pelz's research indicates that a supervisor's ability to exercise influence independently of his superior conditions the reactions of his subordinates to him.[39] Subordinates were favorably disposed to their supervisor only if he engaged in "good" supervisory practices *and also* had enough autonomy from his superior to exercise effective power over the workers' environment. The absence of the latter neutralized the advantages of good supervisory practices.

To explore the significance of hierarchical independence, an index that measures the difference between the supervisor's approach and that of his superior was constructed in City Agency, as previously described. The effects of hierarchical independence differed decidedly from those of authoritarianism.[40] Independent supervisors had more loyal subordinates: four of the five independent supervisors commanded high loyalty in their work group, while only one of the seven others did.[41] It will be recalled that authoritarianism was not associated with loyalty. And whereas authoritarianism was inversely associated

[39] Donald C. Pelz, "Influence: A Key to Effective Leadership in the First-Line Supervisor," *Personnel,* 29 (1952), pp. 209–217.

[40] Despite the fact that there was a slight inverse relationship between the two variables: only one of the five independent supervisors was authoritarian, but five of the seven others were.

[41] Of the twelve supervisors in City Agency, five commanded a high degree of loyalty among their subordinates, two, a medium degree, and five, a low degree.

with work satisfaction and with a service orientation among subordinates, hierarchical independence was not associated with either. In contrast, a willingness to delegate more responsibility to clients was not affected by authoritarianism but was affected by independence. Among workers under independent supervisors, 63 per cent (of 24) favored delegating responsibility to clients by making the clothing allowance a part of the regular budget, whereas only 36 per cent of the (36) workers under more dependent supervisors favored this change in procedure. Perhaps the crucial factor here is the willingness to change existing procedures, which might be encouraged by supervisory practices that demonstrate some freedom from the constraints exerted by superiors in the organization.

Hierarchical independence was also found to promote group solidarity, whereas authoritarianism did not. Specifically, work groups under independent supervisors were more likely than other groups to be characterized by high perceived reciprocity in consultation (this measure was discussed in Chapter V). Thus, under independent supervisors, 32 per cent of 81 consulting relationships mentioned by workers were perceived as reciprocal, and under dependent supervisors only 20 per cent of 101 consulting relationships were so perceived.[42] Cohesive ties among workers, as measured by the proportion of ingroup choices, were also slightly more prevalent under independent supervisors.

These findings may be interpreted by suggesting that the value independent supervisors place on autonomy and self-direction motivates them to differ from their own superiors, on the one hand, and to refrain from interfering in the social interaction among subordinates, on the other. Moreover, supervisors who value independence might manifest this attitude not only in their relations to superiors but also in demands that their subordinates make decisions independently, thus constraining subordinates to obtain advice and social support from one another by establishing consultation partnerships and closer social bonds in general. Finally, work-group solidarity develops perhaps most readily in groups that are somewhat protected against adverse influences of the environment, and supervisors who feel free to differ from their superiors are more likely to furnish such protection than others. For example, conflicts between caseworkers and members of the audit department were frequent in City Agency; all independent supervisors, but only two of the six others, regularly backed their subordinates in

[42] The self-confidence of the supervisor was also found positively associated with the proportion of reciprocal consultations in his work group. Under the eight self-confident supervisors, 32 per cent of all (124) consultation ties were perceived as reciprocal; under the four supervisors lacking self-confidence only 10 per cent (of 58) were so perceived.

these conflicts. It appears, then, that the qualities associated with independence in a supervisor promote group solidarity, in part by facilitating the development of reciprocal bonds in work groups, and in part by creating a challenge that subordinates can best meet by establishing such social ties.[43]

Effects of Consistency. Turning now to another dimension of the supervisor's role, we will consider the effects of consistency in role performance on subordinates. Although there are many conflicting results reported in the literature dealing with the role of parent as it influences the development of the child, one of the most recurrent pleas is for consistency of parental behavior, some authors going so far as to insist that consistently "bad" practices are preferable to erratic behavior. We are not suggesting that the supervisor is a parent-figure for his workers but only that consistency in persons occupying authority positions may well be of generic significance. Lack of consistency in supervision and a lack of clarity in defining the duties of subordinates probably have adverse effects on leadership and on the performance of subordinates.

An experiment by Cohen deals with the effects on worker performance of an ambiguous definition of the task and inconsistent directives from a power figure.[44] Female employees of a public utility company were placed individually in a room with an experimental assistant who was identified as coming from the supervisory level directly above the subject. The power figure varied the consistency of his suggestions from subject to subject; also varied was the clarity of the task assigned. Ambiguous situations and inconsistent directives were found to raise the subject's anxiety, to create less favorable attitudes toward the power figure, and to lower productivity. The field studies of supervisory practices by Comrey and his colleagues also found a positive relationship between consistency and effectiveness of supervision.[45]

Consistency of supervisory performance should be reflected in the extent of agreement among workers when asked about the characteristics of the supervisor. If supervisory behavior varies from time to time and from worker to worker, we would expect workers to manifest little consensus when describing their supervisor; conversely, consistency should find expression in greater consensus. Following this reasoning, a measure of the consistency of role performance of each supervisor in City Agency was obtained by ascertaining the degree of consensus among his subordinates when asked about seven different aspects of his

43 For an intensive study of variations in group structure resulting from differences in supervision based on data collected in City Agency, see Philip M. Marcus, "Expressive and Instrumental Groups," *American Journal of Sociology,* 66 (1960), pp. 54–59.

44 Arthur R. Cohen, "Situational Structure, Self-Esteem, and Threat-Oriented Reactions to Power," Cartwright, *op. cit.,* pp. 35–52.

45 Comrey *et al., op. cit.,* p. 54.

behavior.[46] Role consistency, so measured, was found to be positively associated with worker loyalty to the supervisor. All but one of five consistent supervisors, and only one of the seven less consistent ones, commanded high loyalty in their work groups. The supervisor's role consistency also promoted group solidarity, there being a pronounced relationship between it and perceived reciprocity in consultation. Under the consistent supervisors, 34 per cent of the (88) consultation relationships were perceived as being reciprocal, but under the inconsistent supervisors only 17 per cent (of 94) were so perceived. These findings suggest that stability of supervisory practice promoted the loyalty of subordinates and the development of solidary relationships among them, although they might be interpreted alternatively as resulting from the influence of solidarity and loyalty on consensus about the supervisor.

Consistency of supervisory role performance also appears to account for another set of findings concerning the relationship between the supervisor's self-confidence and the allegiance of his subordinates in City Agency. Table 14 presents these data.[47] The more self-confident supervisors were somewhat more apt to command the loyalty of subordinates if they were strict, but the less self-confident supervisors were

TABLE 14. SUPERVISOR'S SELF-CONFIDENCE, SUPERVISOR'S APPROACH, AND WORKER LOYALTY

Worker Loyalty to Supervisor	*Supervisor's Self-Confidence*			
	High		*Medium or Low*	
	Supervisor's Approach		*Supervisor's Approach*	
	Strict	*Easy*	*Strict*	*Easy*
High	77%	53%	27%	87%
N (= 100%)	13	17	15	15

[46] The seven items on which worker opinion was solicited related to the supervisor's procedure orientation, knowledge of procedures, close supervision, social distance from subordinates, excitability, strictness, and self-confidence. Agreements (the more frequent response) among subordinates were summed for each item (counting one-half of the "don't know" responses) and then divided by seven times the number of subordinates to yield the index of consensus. This index was corrected for size of group, since consensus is more likely to occur by chance in smaller groups. No such measure was obtained in County Agency.

[47] Most supervisors in City Agency were considered self-confident by their subordinates. In order to have enough cases for the analysis in this table, supervisors were divided by the median. This division required that even supervisors who were considered to lack self-confidence by only one worker be classified in the "medium or low" category. (There were two such cases among the twelve supervisors.)